



Sadguru Balumama Shikshan Prasarak Mandal's

K.P.PATIL INSTITUTE OF TECHNOLOGY

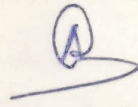
(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary(2025-26)

Department Of Artificial Intelligence & Machine Learning Engineering

Program Code : AI5K
Name of the Student : Sanika Dhangaji Chaugale
Name of the Mentor (Faculty) : Mrs. P. M. Kumbhar
Name of the Mentor (Industry) : Mr. Abhijit Gatade
Enrollment Number : 23213500044

Week	Day & Date	Discussion Topics/Activity	Details of Work Allotted Till Next Session /Corrections Suggested/Faculty Remarks	Signature of Industry Mentor
Week <u>7</u>	Mon, Date <u>14/07/25</u>	◆ Dataset Loading & Data Cleaning	imported data and removed missing /duplicate rows	 
	Tue, Date <u>15/07/25</u>	◆ EDA and Visualization Insights	plotted graphs to explore data relationships	
	Wed, Date <u>16/07/25</u>	◆ Model Selection and feature Encoding	Trained models and handled categorical features	
	Thu, Date <u>17/07/25</u>	◆ Model Tuning & pipeline Creation	imported accuracy with Grid Search CV pipeline	
	Fri, Date <u>18/07/25</u>	◆ Streamlit App Development Started	Designed UI & Connected Prediction model	
	Sat, Date <u>19/07/25</u>	◆ Final Touches & Documentation	Validated inputs and wrote README file	



Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary (2025-26)

Week No: 07

Day: Monday

Date: 14/07/25

"Dataset Loading & Data Cleaning"

Today, I began by loading the car price dataset into the working environment using pandas. I inspected the dataset structure to understand the available features and target variable.

I checked for missing values, duplicate entries, and incorrect data types. After analysis, I removed rows with missing values and dropped duplicate records to ensure data quality. I also standardized column names for consistency and converted appropriate features to correct data types.

This cleaning step helped ensure a solid foundation for further processing and model building.



Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary (2025-26)

Week No: 07

Day: Tuesday

Date: 15/07/25

"EDA & Visualization Insights"

I performed Exploratory Data Analysis (EDA) to understand the relationships between variables. Using Matplotlib and Seaborn, I plotted graphs such as histograms, box plots, heatmaps, and scatter plots to visualize trends and correlations.

I identified which numerical features affected the car price the most and how Categorical features like brand or fuel type influenced the outcome. These insights guided features selection for model building. I also noted any outliers and began thinking about feature transformation if required.



Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary (2025-26)

Week No: 07

Day: Wednesday

Date: 16/07/25

"Model Selection & Feature Encoding"

On day, I began working on converting categorical data into numerical form using techniques like one-hot-encoding and Label Encoding. I split the data into training and test sets. I experimented with multiple regression models such as Linear Regression, Random forest Regressor, and XGBoost.

I compared their initial performances using default parameters. I also evaluated models using metrics like R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error. This step helped shortlist the most promising model for further tuning.



Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary (2025-26)

Week No: 07

Day: Thursday

Date: 17/07/25

"Model Tuning & Pipeline Creation"

Today's was focused on improving model performance. I used Grid Search - CV to perform hyperparameter tuning on the selected models. By defining a parameter grid and applying cross-validation, I found the optimal parameters that improved prediction accuracy. I then created a scikit-learn pipeline to automate the entire process - from preprocessing (scaling, encoding) to prediction. This made the workflow modular, clean, and reproducible. Final models were saved using joblib for future use.



Sadguru Balumama Shikshan Prasarak Mandal's

K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary (2025-26)

Week No: 07

Day: Friday

Date: 18/07/25

"Streamlit App Development Started"

I initiated the development of the web-based user interface using streamlit. The app was designed to take user input like car features mileage, brand, fuel type, and output the predicted car price. I linked the trained model into the backend of the app and added input widgets to collect data from users. A clean and simple layout was chosen to keep the interface user-friendly.

I Successfully connected the model and displayed predictions based on real-time input.



Sadguru Balumama Shikshan Prasarak Mandal's
K.P.PATIL INSTITUTE OF TECHNOLOGY

(DTE CODE : 6814) (MSBTE Code : 1661)

Approved by AICTE, DTE Mumbai & Govt. of Maharashtra, Affiliated to MSBTE Mumbai

Internships Daily Diary (2025-26)

Week No: 07

Day: Saturday

Date: 19/07/25

"Final Touches & Documentation"

On the final day of the week, I completed the last round of testing and improvements. I Validated the app inputs to ensure proper data types and value ranges. The README file was written to provide a clear overview of the project, including the problem statement, steps followed, model accuracy, libraries used, and how to run the app. I also organized the project folder structure and pushed the final version to Github.

All tasks for the car price prediction project were successfully Completed.